Workflow lab using the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Express a causal scenario as a directed acyclic graph (DAG).
- Identify confounders, mediators, and colliders from a DAG.
- Use `dagitty::adjustmentSets()` to find sufficient adjustment sets.

Prerequisites

Conceptual knowledge of confounding.

Background

A DAG is a set of variables joined by arrows that encode assumed causal directions. The graph distinguishes three kinds of third variable on a path between exposure and outcome: a *confounder* is a common cause; a *mediator* lies on the causal path from exposure to outcome; a *collider* is a common effect of two variables on the path. The practical upshot is that adjusting for a confounder reduces bias, adjusting for a mediator removes part of the effect you are trying to estimate, and adjusting for a collider *creates* bias where none existed.

`dagitty` and `ggdag` let you declare a DAG in text, visualise it, and then query it for adjustment sets — the minimal variable sets that block all back-door paths from exposure to outcome without opening new ones. The right adjustment set depends on the question. For the total effect of X on Y, you want to close all back-door paths but leave mediators alone; for the direct effect, you also block mediators.

M-bias is the classic example of collider adjustment: conditioning on a variable that is a common effect of an unmeasured cause of X and an unmeasured cause of Y opens a path and biases the estimate. This is why "adjust for everything" is bad advice.

Setup

```r
library(tidyverse)
library(dagitty)
library(ggdag)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Write a confounder, mediator, and collider scenario as DAGs; ask `dagitty` which variables to adjust for.

2. Approach

```r
  X -> Y
  C -> X
  C -> Y
  X [exposure]
  Y [outcome]
}')

dag2 <- dagitty('dag {
  X -> M -> Y
  X -> Y
  X [exposure]
```

```
  Y [outcome]
}')

dag3 <- dagitty('dag {
  X -> Z
  Y -> Z
  X -> Y
  X [exposure]
  Y [outcome]
}')

ggdag(dag1) + theme_dag()
ggdag(dag2) + theme_dag()
ggdag(dag3) + theme_dag()
```

3. Execution

```
               effect = "total")
```

```
adjustmentSets(dag2, exposure = "X", outcome = "Y",
               effect = "total")
adjustmentSets(dag2, exposure = "X", outcome = "Y",
               effect = "direct")
```

```
adjustmentSets(dag3, exposure = "X", outcome = "Y")
```

4. Check

Simulate dag1 and verify that adjusting for C recovers the true effect while failing to adjust biases it.

```
c_ <- rnorm(n)
x  <- 0.6 * c_ + rnorm(n)
y  <- 0.4 * x + 0.7 * c_ + rnorm(n)
df <- tibble(c_, x, y)

coef(lm(y ~ x,        data = df))["x"]
coef(lm(y ~ x + c_,   data = df))["x"]
```

The unadjusted coefficient is inflated; the adjusted one is close to 0.4 as simulated.

5. Report

> A directed acyclic graph is a compact, testable statement of causal assumptions. Dagitty identified the confounder C as the required adjustment set in the first DAG; the collider Z in the third DAG is explicitly *not* in any adjustment set, and conditioning on it would introduce selection bias.

Common pitfalls

- Listing "every variable we measured" as the adjustment set.
- Adjusting for a mediator and calling the result a total effect.
- Using automated variable selection on observational data without a DAG.
- Drawing the DAG after the analysis.

Further reading

- Textor J et al. (2016), *Robust causal inference using directed acyclic graphs: the R package 'dagitty'.*
- Hernán MA, Robins JM. *Causal Inference: What If.*
- Greenland S, Pearl J, Robins JM (1999), *Causal diagrams for epidemiologic research.*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)